

Exercise 0

A guided first journey from source code to a running program

Introduction to Computer Science

This exercise is not graded, and you are not expected to know the tools in advance. Its purpose is to let you install everything calmly, try the complete course workflow once, and solve any technical problems before the graded exercises begin. At the end, the submission system will send you an email with a grade of **100**. In this exercise, that number simply means that your setup and submission were successful.

During the exercise, we will move through the same journey that you will follow in later assignments. We will begin with code stored on GitHub, bring a copy to your computer, compile and run it on your local computer and on your Arduino board. Finally, you will connect to the departmental server, compile and run the code there, and submit it. More specifically:

1. You will install Git, VS Code, GCC, and the Arduino command-line tools.
2. You will create your own fork of Exercise 0 and clone it to Linux.
3. You will compile and run C code on your local machine.
4. You will compile and upload a program to an Arduino Uno, and then communicate with it through the serial interface.
5. You will connect to the departmental Linux server and test the program there.
6. You will prepare the required ZIP archive, submit it, and understand the feedback returned by the submission system.

1 Before you begin

Before we start, make sure you have your departmental username and password. You can find your username [here](#), under the entry UNIX-MACS or ESC. Your password is the same password you use to log in to In-Bar.

You will also need a GitHub account. GitHub is the website that we use to distribute the exercise files and keep track of code. If you do not have an account yet, create a free one at github.com. By the way, GitHub offers a Student Developer Pack with various benefits for students. You are welcome to apply for it in your free time (:

We also recommend creating one simple folder for all course work, such as `~/intro2cs`. Keeping the path short and using English letters prevents several surprisingly common problems with compilers and file-transfer tools. For the same reason, avoid placing the course folder inside OneDrive, Google Drive, or another synchronized folder.

2 Install the Linux tools

2.1 Install Git

Git is a tool for managing the files in a programming project and keeping track of changes over time. GitHub stores Git projects, called repositories, while Git is the program on your computer that lets you copy and work with them.

The commands in this guide assume Ubuntu or another Debian-based Linux distribution. Open the Terminal and install Git with:

```
sudo apt update
sudo apt install git
```

Linux may ask for your password. Nothing appears while you type it; this is normal. Press Enter when you finish, confirm the installation if asked, and then check that Git is available:

```
git --version
```

You should see a line beginning with `git version`. The exact version number is not important.

2.2 Install an IDE

An IDE (Integrated Development Environment) is a program that helps you write and develop code. It usually combines a code editor with useful tools such as syntax highlighting, error detection, debugging, and running your program. In this course, the official IDE is Visual Studio Code (VS Code). You may use another IDE, such as [CLion](#), if you prefer. However, the official course IDE is VS Code, so demonstrations, screenshots, and technical support will generally assume that you are using it. Regardless of the editor you choose, all exercises must compile and run using the tools defined by the course. Most importantly: The C compiler used in this course is GCC. Your code must compile correctly using GCC. It is important to distinguish the editor from the compiler: VS Code edits the file, while GCC translates that file into a program the computer can execute.

1. Open the official [VS Code Linux installation guide](#) and choose the package for your distribution. On Ubuntu, download the `.deb` package.

2. Open the downloaded package with the Software Center, or install it from the Terminal with `sudo apt install ./FILE.deb`, replacing `FILE.deb` with the downloaded filename.
3. Start VS Code. Select the Extensions icon on the left, search for **C/C++**, and install the extension whose publisher is Microsoft. This extension teaches VS Code how to understand C source files.

2.3 Install GCC

In this course, we use the GCC compiler. A compiler translates human-readable C code into a machine-language executable that the computer can run. On Ubuntu and Debian-based systems, the `build-essential` package installs GCC and the other basic compilation tools that we need.

1. Open the Terminal.
2. Update the package list and install the development tools:

```
sudo apt update
sudo apt install build-essential
```

The package manager places GCC on the normal command path automatically, so no separate PATH configuration is needed.

Let us now ask each tool for its version. These commands do not modify anything; they simply confirm that Linux can find the programs:

```
git --version
gcc --version
ar --version
```

Checkpoint

Each command prints a version and none prints “is not recognized.”

2.4 Disable AI assistance

AI tools are becoming an important part of software development, and you will probably use them extensively in the future. During this course, however, we want you to disable them while working on the exercises.

The purpose of an introductory programming course is not only to learn C syntax. More importantly, it is to develop **logical thinking, algorithmic thinking, problem-solving skills, and the ability to debug your own programs.**

AI code generation can be a bit of a **honey pot** at this stage. When you are stuck, it is extremely tempting to let the AI finish a function, fix a bug, or generate an algorithm for you. Unfortunately, the struggle of figuring out these things yourself is exactly where much of the learning happens.

Therefore, please disable VS Code’s AI features while working on the course exercises:

1. Open VS Code’s Settings by selecting `File > Preferences > Settings`. You can also press `Ctrl+,`.
2. Enter `chat.disableAIFeatures` in the Settings search box.

3. Find Chat: Disable AI Features and enable its checkbox.

This setting disables and hides VS Code's built-in AI functionality, including AI chat, Copilot, and AI-generated inline suggestions.

You should still keep the **Microsoft C/C++ extension** installed. Its IntelliSense features, such as completing variable and function names and displaying function parameters, are regular programming-editor features and are allowed.

Important

Getting stuck is part of the exercise. Read the error, debug the program, think about the algorithm, ask questions, and try again. These are exactly the skills this course is designed to develop.

3 Fork and clone Exercise 0

The course staff keep the original exercise in a GitHub *repository*. We do not work directly inside that original repository. Instead, you will create a *fork*, which is a GitHub copy associated with your account. You will then *clone* the fork, which downloads a working copy to your computer. This may sound like two copies of the same thing, and that is exactly right: one copy lives on GitHub, while the other is the folder in which you actually edit and run code.

1. Sign in to GitHub and open the course account: github.com/intro2cs-biu.
2. Open the ex0 repository.

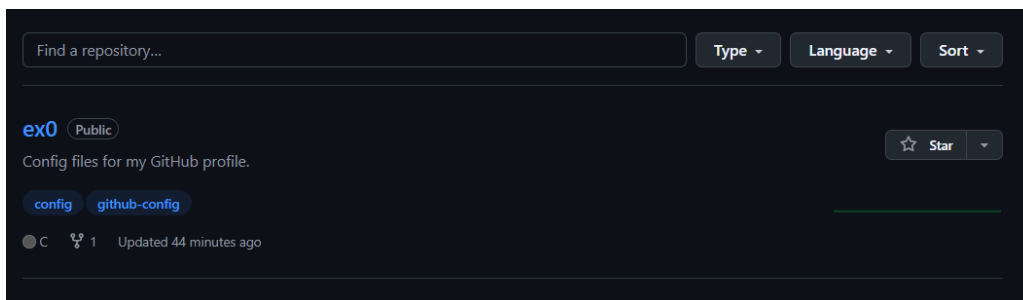


Figure 1: Open the Exercise 0 repository from the course organization. The organization name and GitHub layout may differ slightly from this screenshot.

3. Click Fork, choose your account as owner, and create the fork.

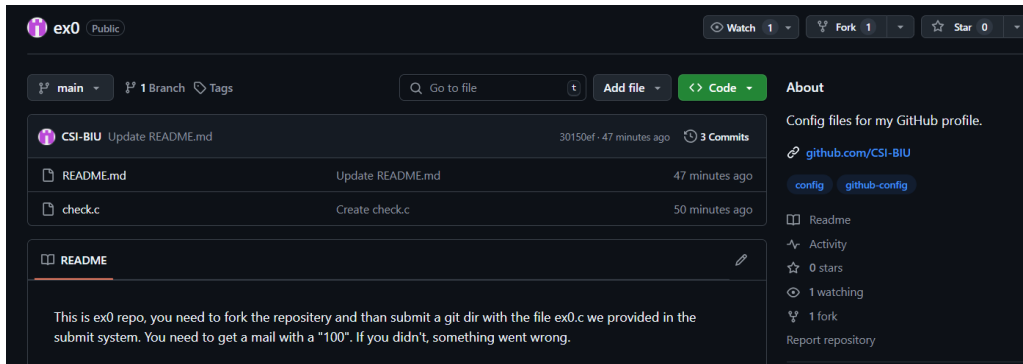


Figure 2: The Fork button is at the upper right of the repository page.

Important

Never publish solutions in a public repository. Your fork must be private if GitHub permits changing its visibility, and solution files must not be shared with other students.

4. In your fork, click the green Code button, choose HTTPS, and copy the URL. It should resemble `https://github.com/YOUR-USER-NAME/ex0.git`.

Now we will create your course folder and download the repository. Open the Terminal and enter the commands below one line at a time. In the `git clone` line, paste the address that you copied from your own fork:

```
mkdir -p ~/intro2cs
cd ~/intro2cs
git clone https://github.com/YOUR-USER-NAME/ex0.git
cd ex0
ls
```

The `mkdir` command creates a directory, `cd` moves into it, and `git clone` downloads the repository. The final `ls` command lists the files so that you can confirm they arrived. If GitHub asks you to sign in, follow its browser sign-in flow.

Checkpoint

The `ls` command shows `ex0.c`, and `git status` says you are inside a working tree.

4 Compile and run the first C program

You now have your first C source file. A source file is still just text, the computer cannot execute it directly. We will open it in VS Code, ask GCC to compile it, and then run the executable that GCC creates.

Open the cloned folder in VS Code by entering:

```
code .
```

The dot means “the current folder.” When VS Code opens, select `ex0.c`. You will see a short program whose job is to print `hello world`. Return to Terminal, make sure you are still in the folder containing `ex0.c`, and compile it with:

```
gcc -std=c99 -Wall -Wextra ex0.c -o ex0
```

Here, `-std=c99` asks GCC to use the version of C required by the course, and `-Wall -Wextra` asks it to warn us about issues in the code. The `-o ex0` part names the program that GCC creates. If the command ends without an error message, compilation succeeded. Now run the new program:

```
./ex0
```

The Terminal should display:

```
hello world
```

Tip

Compilation and execution are different steps. GCC creates `ex0`; typing `./ex0` runs the program from the current directory. Recompile after every source change.

5 Install the Arduino command-line tools

5.1 Install Arduino CLI

Arduino CLI is the official command-line tool for working with Arduino boards. We will use it to detect connected boards, compile and upload programs, and communicate with the board through the serial monitor.

Open the official [Arduino CLI installation page](#). Download the 64-bit Linux archive that matches your processor, extract the `arduino-cli` executable, and move it to a directory on your PATH. For example, from the directory containing the extracted file, run `sudo install arduino-cli /usr/local/bin/arduino-cli`. Then verify the installation and install support for the Arduino Uno:

```
arduino-cli version
arduino-cli core update-index
arduino-cli core install arduino:avr
arduino-cli core list
```

The final command should list `arduino:avr`. This also installs Arduino’s board support and drivers/tools used for an Uno.

5.2 Install the raw AVR compiler and uploader

The processor inside an Arduino Uno is different from the processor inside your computer. Consequently, the normal Linux GCC cannot create a program for the Uno. We use `avr-gcc`, a version of GCC that targets AVR microcontrollers. The program `avrdude` then transfers the compiled result to the board.

On Ubuntu or Debian-based Linux, open the Terminal and run:

```
sudo apt update
```

```
sudo apt install gcc-avr binutils-avr avr-libc avrdude
```

Then open a new Terminal and verify:

```
avr-gcc --version
avr-objcopy --version
avrdude --version
```

6 Download and install the Intro2CS library

The Arduino tools know how to compile a program for the Uno, but they do not yet provide the functions used to control the components in the course kit. The Intro2CS library adds these functions.

The library has an [official course tutorial and documentation](#). It contains the current library download, wiring diagrams, compilation commands, and documentation for each library module.

Use the guide to install the library as follows:

1. Download `intro2cs-lib-linux.zip` library from this [link](#).
2. Move the downloaded ZIP file into your main course folder, `~/intro2cs`. This is the folder that already contains the cloned `ex0` repository.
3. Extract the ZIP with your file manager, or run `unzip intro2cs-lib-linux.zip` in the Terminal. The extracted folder named `intro2cs-lib` should be beside `ex0`, not inside it.

Your main course folder should now look like this:

```
~/intro2cs/
|-- intro2cs-lib/
|   |-- headers/
|   |-- intro2cs-arduino-lib.a
|   '-- intro2cs-arduino-simulator-lib.a
|-- ex0/
|   |-- ex0.c
|   '-- ex0-arduino.c
|-- ex1/
|-- ...
```

The `ex1` line illustrates where later repositories will go, it does not need to exist yet. If you see another `intro2cs-lib` folder inside the first one, move the inner folder up one level.

In future exercises, clone each repository into `~/intro2cs`. The relative path `../intro2cs-lib` will then continue to work without installing the library again.

7 Connect the Arduino circuit

We are now ready to run our first piece of code on the Arduino board. Make sure the usb cable is disconnected from the board before adding or moving wires. This protects the board from an accidental short circuit while you work. Build the circuit shown below using the Arduino Uno, breadboard, LED with its series resistor, and the 16-by-2 LCD. Take your time and follow one wire at a time. Before reconnecting USB, check the LED polarity and verify each power and ground connection.

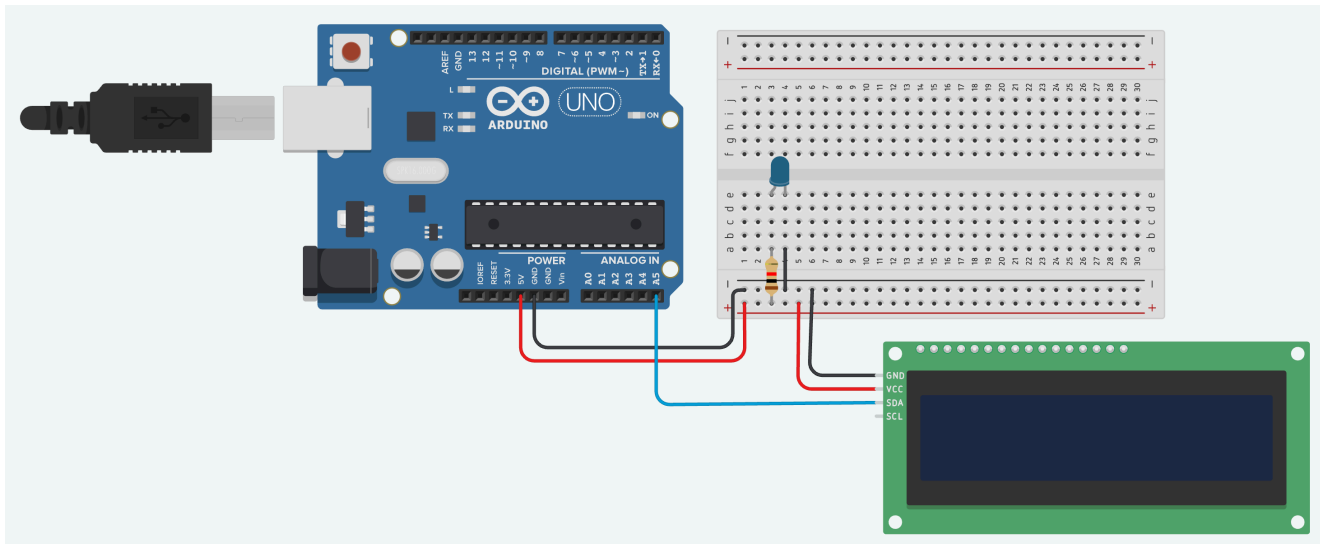


Figure 3: Exercise 0 circuit. Follow the diagram exactly.

Before compiling, open `ex0-arduino.c` in VS Code and take a moment to read through it. You are not expected to understand every detail yet, we will study these ideas more carefully during the course. For now, let us briefly see what each line asks the Arduino to do:

```
#include "arduino.h"
#include "lcd.h"
#include "serial.h"
#include <stdio.h>

int main(void) {
    pinMode(2, OUTPUT);
    lcdBegin(16, 2);
    serialBegin(9600);
    printf("Hello, World!\n");
    lcdPrintf("Hello, World!");
    digitalWrite(2, HIGH);
}
```

The `#include` lines make functions from the course library and the standard C library available to the program. The function `main` is where the program begins running. Inside it, `pinMode(2, OUTPUT)` prepares pin 2 to send an electrical signal, and `lcdBegin(16, 2)` prepares the LCD, which has 16 characters in each of its two rows.

The call `serialBegin(9600)` starts communication with the computer at a speed of 9600 baud. It also waits briefly so that the serial monitor has time to reconnect after the Arduino resets; this helps prevent the beginning of the first message from being lost or joined to output from an earlier run. The following `printf` sends `Hello, World!` to the computer through that serial connection, while `lcdPrintf` displays the same message on the LCD. Finally, `digitalWrite(2, HIGH)` sends a `HIGH` signal through pin 2, which turns on the connected LED.

This quick explanation is enough for now. You will learn what functions, headers, pins, serial communication, and the values `HIGH` and `OUTPUT` mean in much greater detail as the course progresses.

8 Compile and upload to the Arduino Uno

Compiling for the Arduino has two visible stages. First, `avr-gcc` produces an ELF file containing the machine code and useful metadata. Then `avr-objcopy` creates the smaller HEX file that will be uploaded to the board. Run the following commands:

```
avr-gcc -mmcu=atmega328p -DF_CPU=16000000UL -Os -std=c99 \  
-I../intro2cs-lib/headers ex0-arduino.c \  
../intro2cs-lib/intro2cs-arduino-lib.a -o ex0-arduino.elf  
  
avr-objcopy -O ihex -R .eeprom \  
ex0-arduino.elf ex0-arduino.hex
```

If your files are stored in different folders, update the paths in the commands accordingly. In future exercises, you will also need to replace the exercise file names (`ex0-arduino.c`, `ex0-arduino.elf`, and `ex0-arduino.hex`) with the appropriate names for that exercise.

Now connect the Arduino board to your computer using the USB cable. Linux assigns the connected board a serial-port name such as `/dev/ttyACM0`. The number can change when you use a different USB socket, so we should ask Arduino CLI instead of guessing. Run the following command:

```
arduino-cli board list
```

Look for Arduino Uno, FQBN `arduino:avr:uno`, and a port such as `/dev/ttyACM0`. The FQBN is Arduino's precise name for the type of board. To upload the code to the board, replace `/dev/ttyACM0` in the following command with the port shown on your computer:

```
avrdude -p atmega328p -c arduino -P /dev/ttyACM0 -b 115200 -D \  
-U flash:w:ex0-arduino.hex:i
```

Checkpoint

If everything worked correctly, the LCD should display `Hello, World!`, and the LED should be lit.

9 Serial interface tutorial

So far, we have used the computer only to upload a program to the Arduino. Serial communication creates an ongoing connection between the computer and the board, allowing them to exchange information while the program is running. The Arduino board can send text and values to the computer, and the computer can send input back to the Arduino. This makes it especially useful for observing, interacting with, and debugging a running program.

The communication travels through the USB cable. Both sides must agree on a speed, called the *baud rate*, in this exercise that speed is 9600. The starter program contains the following essential pattern:

```
#include "serial.h"  
#include <stdio.h>  
  
int main(void) {  
    serialBegin(9600);  
    printf("Hello, World!\n");  
}
```

```
}
```

The header makes the course serial function available. Calling `serialBegin(9600)` connects the familiar standard C input and output functions to the Arduino serial port. Always call it once before using `printf`, `getchar`, or `scanf`. Ending output with `\n` starts a new line, which makes the terminal readable.

9.1 Open the serial monitor

A serial monitor is a small terminal window connected to the board. First ask Arduino CLI for the current port, then open that port at 9600 baud:

```
arduino-cli board list
arduino-cli monitor -p /dev/ttyACM0 --config baudrate=9600
```

Remember to replace `/dev/ttyACM0` with your port. You should see `Hello, World!`. Press `Ctrl+C` when you want to close the monitor. If Arduino CLI reports a configuration error, ask it to describe the settings supported by this port:

```
arduino-cli monitor -p /dev/ttyACM0 --describe
```

Press the Uno's reset button after opening the monitor if the startup message has already passed. Opening a serial connection can itself reset many Uno boards, so a repeated startup message is normal.

Important

`/dev/ttyACM0` is only an example. Close any serial monitor before re-uploading your code; only one program can normally use the port at a time.

10 Test on the departmental server

The departmental server is a Linux computer maintained by the department. Running your program there confirms that it behaves correctly in the same environment used by the course systems and staff.

Before your first connection, you must set up two-factor authentication:

1. Download and install **Google Authenticator** on your phone ([iPhone](#), [Android](#)).
2. Go to mfar.esc.biu.ac.il and enter your username.
3. After confirming your username, a QR code will be sent to your Bar-Ilan email address. Open Google Authenticator and scan this QR code.
4. Google Authenticator will now display a six-digit verification code that changes every 30 seconds. You will use this code when connecting to the departmental server.

Open a Terminal and connect through the departmental gateway. Replace both occurrences of `username` with your departmental username:

```
ssh -J username@vpn.esc.biu.ac.il:2222 \
    username@planet.cs.biu.ac.il
```

Enter your password and changing verification code when asked. Passwords do not appear while you type. After connecting, prepare a personal folder and then leave the server temporarily:

```
mkdir -p ~/intro2cs/ex0
exit
```

Copy `ex0.c` from your local `ex0` folder to the server:

```
scp -J username@vpn.esc.biu.ac.il:2222 ex0.c \
  username@planet.cs.biu.ac.il:~/intro2cs/ex0/
```

Connect again with the same `ssh` command, then compile and run:

```
cd ~/intro2cs/ex0
ls
gcc -std=c99 -Wall -Wextra ex0.c -o ex0
./ex0
```

The expected output is `hello world`.

10.1 Verify the Arduino compilation on the server

You can also use the departmental server to check that your Arduino source code compiles correctly. This is useful because it tests the build in a standard course environment before you upload the result to the physical board.

Use `scp` with the same jump-host option to copy `ex0-arduino.c` and the `intro2cs-lib` directory to the server. The server should have the same neighboring `ex0` and `intro2cs-lib` folders as your local course directory.

In the remote server terminal, enter:

```
cd ~/intro2cs/ex0
avr-gcc --version
avr-gcc -mmcu=atmega328p -DF_CPU=16000000UL -Os -std=c99 \
  -I../intro2cs-lib/headers ex0-arduino.c \
  ../intro2cs-lib/intro2cs-arduino-lib.a -o ex0-arduino.elf
avr-objcopy -O ihex -R .eeprom \
  ex0-arduino.elf ex0-arduino.hex
```

If these commands finish without an error, the Arduino program compiled successfully. The file that must be uploaded to the board is `ex0-arduino.hex`. Leave the server with `exit`, then copy the file back to your local `~/intro2cs/ex0` folder:

```
scp -J username@vpn.esc.biu.ac.il:2222 \
  username@planet.cs.biu.ac.il:~/intro2cs/ex0/ex0-arduino.hex .
```

The Arduino is connected by USB to your Linux computer, not to the remote departmental server. For that reason, the remote server normally cannot upload directly to the board through SSH. After downloading the HEX file, close the serial monitor and upload it from a local Terminal, replacing `/dev/ttyACM0` with the port shown on your computer:

```
arduino-cli board list
```

```
avrdude -p atmega328p -c arduino -P /dev/ttyACM0 -b 115200 -D \
-U flash:w:ex0-arduino.hex:i
```

10.2 Getting More Comfortable with Linux

If you would like to become more comfortable with the Linux environment and the commands used above, we recommend working through [this introductory Linux tutorial](#). You do not need to complete the entire tutorial for Exercise 0, but its sections about navigating directories, listing files, and working with the command line are useful throughout the course.

11 Compare your output with the submission simulator

When you submit an exercise, the submission system runs basic automatic tests. These tests do not connect your program to a physical Arduino. Instead, they use the course simulator library, which replaces hardware actions with text. For example, an instruction that sets a pin to HIGH produces a line of text describing that action. The tests compare the program's output with the output expected for the exercise.

This is the reason you may want to run the simulator on your own computer and on the departmental server: it allows you to see exactly what the basic automatic tests will see. It is not a replacement for using the Arduino board. We use it only to compare textual output before submitting.

Compile the same `ex0-arduino.c` source file with the normal Linux `gcc` compiler, but link it with `intro2cs-arduino-simulator-lib.a` instead of `intro2cs-arduino-lib.a`. The simulator library implements the same course functions while printing their effects in the terminal.

Open Terminal in `~/intro2cs/ex0` or in the departmental server and enter:

```
gcc -std=c99 -Wall -Wextra -I../intro2cs-lib/headers \
ex0-arduino.c \
../intro2cs-lib/intro2cs-arduino-simulator-lib.a \
-o ex0-simulator

./ex0-simulator
```

Compare this with the physical-Arduino compile command used earlier. The physical build uses `avr-gcc` and `intro2cs-arduino-lib.a`, the simulated build uses the ordinary `gcc` and `intro2cs-arduino-simulator-lib.a`. Your C source and header includes remain unchanged.

For the starter program, the simulator stays silent while pin 2, the LCD, and serial communication are initialized. Its output begins only when the program performs a visible output operation: it prints `Hello, World!`, renders the LCD update, and indicates that pin 2 was set to HIGH. Compare the order, spelling, capitalization, spaces, and numeric values carefully. Automated comparisons are exact, so output that looks equivalent to a person may still differ because of one extra character or a missing newline.

Important

Passing this comparison only shows that the program produced the expected basic simulator output. You must still compile and run the program on the real Arduino.

12 Prepare the ZIP submission

Open a Terminal in the `ex0` repository. Make sure that `ex0.c` and `ex0-arduino.c` are present, then create the ZIP with your file manager's **Compress** action or with:

```
zip -r ex0.zip . -x '*.o' '*.elf' '*.hex' 'ex0' \
    'ex0-simulator' 'ex0.zip'
```

If `zip` is not installed, run `sudo apt install zip`. Open the archive before submitting it and confirm that the two C files are present. Do not include generated programs or build files unless the submission instructions explicitly request them.

13 Submit and verify the result

You have now reached the final step. Open submit.cs.biu.ac.il. Sign in with your departmental username, password, and verification code. Choose course **89-110**, group **01**, and Exercise **0**, choose the ZIP, send it, and confirm.

The form is titled "שלח/קבצים (2)" and "בחר/תרגיל (1)". It contains the following elements:

- Section (1) - בחר/תרגיל:** Three dropdown menus for "קורס:" (89-110), "קבוצה:" (01), and "תרגיל:" (ex_0).
- Section (2) - שלח/קבצים:** A text input field for the ZIP file name, a "בחר" button, and "מחק" and "שלח" buttons. Below is a "קבצים שהתקבלו:" section.
- Section (3) - אשר/ההגשה:** A red warning message "ללא אישור, התרגיל לא נחשב כהוגש" and three buttons: "חזרה", "ביטול", and "אישור".

Figure 4: Select the course, group, exercise, and ZIP before confirming.

After submission, the system examines the archive and sends feedback, normally to your university email address. A successful report ends with a grade of 100.

Checkpoint

Do not consider Exercise 0 complete until the feedback email shows **100**. If the grade is lower, read the report, correct the named issue, rebuild the ZIP, and submit again.

14 If something does not work

Technical setup problems are normal, especially the first time several tools must work together. An error message is information, not a sign that you are bad at programming. Read the first relevant error line,

```

The following files have been accepted for assignment "ex0":

.git/config (292 bytes)
.git/description (73 bytes)
.git/HEAD (21 bytes)
.git/hooks/applypatch-msg.sample (478 bytes)
.git/hooks/commit-msg.sample (896 bytes)
.git/hooks/fsmonitor-watchman.sample (4726 bytes)
.git/hooks/post-update.sample (189 bytes)
.git/hooks/pre-applypatch.sample (424 bytes)
.git/hooks/pre-commit.sample (1643 bytes)
.git/hooks/pre-merge-commit.sample (416 bytes)
.git/hooks/pre-push.sample (1374 bytes)
.git/hooks/pre-rebase.sample (4898 bytes)
.git/hooks/pre-receive.sample (544 bytes)
.git/hooks/prepare-commit-msg.sample (1492 bytes)
.git/hooks/push-to-checkout.sample (2783 bytes)
.git/hooks/update.sample (3650 bytes)
.git/index (209 bytes)
.git/info/exclude (240 bytes)
.git/logs/HEAD (179 bytes)
.git/logs/refs/heads/main (179 bytes)
.git/logs/refs/remotes/origin/HEAD (179 bytes)
.git/objects/pack/pack-e552c6258960418a1ff61dc95326756f0aa137ef.idx (1324 bytes)
.git/objects/pack/pack-e552c6258960418a1ff61dc95326756f0aa137ef.pack (3084 bytes)
.git/packed-refs (112 bytes)
.git/refs/heads/main (41 bytes)
.git/refs/remotes/origin/HEAD (30 bytes)
ex0.c (78 bytes)
README.md (207 bytes)

Your grade for assignment "ex0" is "100".

"ex0" was checked with 1 inputs:

(1) 0 line errors, 100 points out of 100.
Input
Correct Output
hello world
Your Output
hello world

Submitted: Mon, 28 Oct 2024, 1:05.
Last submission date with no reduction points: 28 Oct 2024.

```

Figure 5: Example of successful Exercise 0 feedback.

find the matching case below, and check one possible cause at a time.

gcc or git is not recognized

Run `sudo apt update` and reinstall the relevant package. Programs installed by APT are normally placed on the command path automatically.

Header file such as `arduino.h` is missing

Run the command from `ex0` and verify that `..\intro2cs-lib\headers` exists. Do not change include lines to absolute paths.

No board appears

Try another data-capable USB cable or port, reconnect the board, and rerun `arduino-cli board list`. If the board is visible but access is denied, add your user to the serial-port group with `sudo usermod -a -G dialout $USER`, then sign out and back in.

Upload cannot open the serial port

Close the serial monitor and any Arduino IDE window, confirm the port again, and retry. Replace the example `/dev/ttyACM0` path everywhere.

No serial text

Set the monitor to 9600 baud, open it, and press Reset. Confirm that the code calls `serialBegin(9600)` before printing.

Support resources: [faculty support](#), [submission troubleshooting](#), and [helpdesk](#). Consult classmates and the course forum before opening a helpdesk request.

15 Before you finish

Take one final minute to review what you accomplished. Every statement below should now be true. If one is not, return to that section before moving on to a graded exercise.

- ☐ I can open Git, GCC, and VS Code on my Linux computer.
- ☐ I created both a GitHub fork and a local clone of Exercise 0, and I made sure that I am not publishing solutions publicly.
- ☐ I compiled `ex0.c` locally and saw it print `hello world`.
- ☐ I ran the program on the real Arduino and saw its text in the serial monitor.
- ☐ I compared the program's textual output with the output produced by the submission simulator.
- ☐ I compiled and ran `ex0.c` on `planet.cs.biu.ac.il`.
- ☐ I opened the ZIP and confirmed that `ex0.c` and `ex0-arduino.c` are both at its top level.
- ☐ I received a submission feedback email showing **100**.